Mathematical Analysis 2

Metric spaces and continuity: exam

Time: 45 minutes Total: 10 points

Name: _____ **Score:** _____ / 10

Written or printed materials are allowed. Electronic devices are not allowed. Work independently and write your solutions on separate paper. Justify all answers. You may cite results from the course, but explain why their hypotheses apply. Unless another metric is specified, use the Euclidean metric.

1. A weighted metric [5 points]

For $x = (x_1, x_2)$ and $y = (y_1, y_2)$ in $\mathbb{R}^2$, define

$$\rho(x, y) = |x_1 - y_1| + 2\,|x_2 - y_2|\,.$$

(a) Prove that ρ is a metric on $\mathbb{R}^2$. [3 points]

(b) Describe and sketch the open ball $B_\rho((1, -1), 1)$. Give the vertices of its boundary and state whether the boundary is included. [2 points]

2. A function of two variables [5 points]

Consider the real-valued function

$$F(x, y) = \sqrt{\ln(4 - x^2 - y^2)}.$$

(a) Find its domain D. Decide separately whether D is open and whether D is closed in $\mathbb{R}^2$, and justify both answers. [3 points]

(b) Determine where $F : D \to \mathbb{R}$ is continuous, including at boundary points of D. [2 points]

Solutions and marking guide

Suggested timing: Problem 1: 20 minutes; Problem 2: 20 minutes; final review: 5 minutes. Each problem is worth 5 points, for a total of 10. Accept any mathematically correct alternative argument and award partial credit for correct reasoning. The point allocations below are a guide.

1. A weighted metric

(a) Metric axioms. The value $\rho(x, y)$ is finite and nonnegative. Both summands are nonnegative, so $\rho(x, y) = 0$ holds exactly when $x_1 = y_1$ and $x_2 = y_2$, that is, when $x = y$. Symmetry follows from $|x_i - y_i| = |y_i - x_i|$. For any $x, y, z \in \mathbb{R}^2$, the triangle inequality for absolute value gives

$$\begin{aligned}\rho(x, y) &= |x_1 - y_1| + 2\,|x_2 - y_2| \\ &\le |x_1 - z_1| + |z_1 - y_1| + 2\,|x_2 - z_2| + 2\,|z_2 - y_2| \\ &= \rho(x, z) + \rho(z, y).\end{aligned}$$

Thus all metric axioms hold.

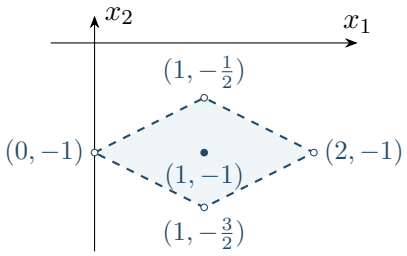
Marking: 1 point for nonnegativity, the zero-distance condition, and symmetry; 2 points for a valid triangle-inequality argument.

(b) The open ball. Substitution of the centre gives

$$\boxed{B_\rho((1, -1), 1) = \{(x_1, x_2) : |x_1 - 1| + 2\,|x_2 + 1| < 1\}.}$$

It is the interior of a diamond centred at $(1, -1)$, with horizontal half-width 1 and vertical half-height $1/2$. The boundary has vertices

$$(0, -1), \quad (2, -1), \quad (1, -\tfrac{1}{2}), \quad (1, -\tfrac{3}{2}).$$



All boundary edges and vertices are excluded because the inequality is strict.

Marking: 1 point for a correct description of the diamond and its vertices; 1 point for an accurate sketch showing that the boundary is excluded. A correctly labelled sketch may also supply the description.

2. A function of two variables

(a) Domain and its topology. The logarithm requires $4 - x^2 - y^2 > 0$, while the outer square root requires $\ln(4 - x^2 - y^2) \geq 0$. Combining these conditions gives

$$4 - x^2 - y^2 \geq 1, \qquad \text{hence} \qquad \boxed{D = \{(x, y) : x^2 + y^2 \leq 3\}.}$$

This is the closed disk of radius $\sqrt{3}$ centred at the origin. It is closed as the preimage of $(-\infty, 3]$ under the continuous polynomial $(x, y) \mapsto x^2 + y^2$. It is not open: $(\sqrt{3}, 0) \in D$, but every ball about this point contains a point $(\sqrt{3} + t, 0) \notin D$ with $t > 0$.

Marking: 1 point for the correct domain; 1 point each for closedness and failure of openness, with justification. Merely requiring the logarithm's argument to be positive is insufficient.

(b) Continuity. On D, the polynomial $4 - x^2 - y^2$ takes values in $[1, 4]$. The logarithm is continuous there and takes values in $[0, \ln 4]$. The square root is continuous on $[0, \infty)$, including at 0. Thus F is a composition of continuous functions and is **continuous at every point of** D, including its boundary. Continuity at a boundary point considers only points of the domain.

Marking: 1 point for the conclusion, including the boundary; 1 point for a valid composition argument with the relevant domains.